

Computational Fluid Dynamics for Aerospace Applications

Lecture 8 Verification and Validation

Verification and Validation

- Verification
- Convergence
- Conservation
- Validation

Verification and Validation

- A CFD result that looks reasonable can still be wrong
 - We can assess this with verification and validation

Verification

Solving the
equations correctly

Validation

Solving the correct
equations

- This is a sequential process: verify before you validate
- Check that the code is implemented correctly before checking whether it gives you a good answer

Verification

- Code verification: Is the numerical method implemented well?
- Check for bugs, wrong signs, indexing errors, etc.
- Stability: Tendency of our approximations to remain bounded
 - We have performed numerical analysis to reveal the required relationships between Δt and Δx to ensure a given numerical scheme remains stable
- Convergence: Tendency of our approximations to approach the exact solution as Δt and Δx approach zero
 - We have performed numerical analysis to characterize how quickly different schemes are supposed to do this
- Conservation: Tendency of our schemes to conserve quantities

Convergence

- New codes/features usually have their convergence verified against the expected convergence:
- Numerically solve a case that has an exact analytical solution that you can compare against
- Measure the difference between the true solution and your approximation
- Repeat for different values of Δx and Δt to ensure that error shrinks at the expected rate

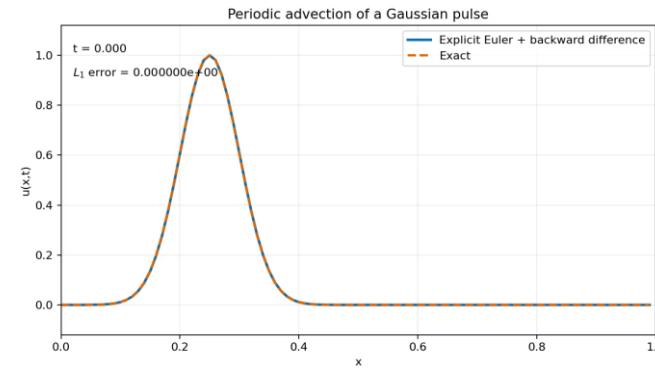
Convergence

Note you (will) do this
yourself on assignment #1

- Example, consider linear advection: $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$
- We know the exact solution for this equation is given by $u(x, t) = u_0(x - at)$, which describes our initial solution advected at velocity a
- Pick the scheme you're using, e.g., EE FO BD
 - $u_j^{n+1} = u_j^n - \frac{a\Delta t}{\Delta x} (u_j^n - u_{j-1}^n) + O(\Delta t, \Delta x)$
- Introduce some error definition, e.g., the L1 norm

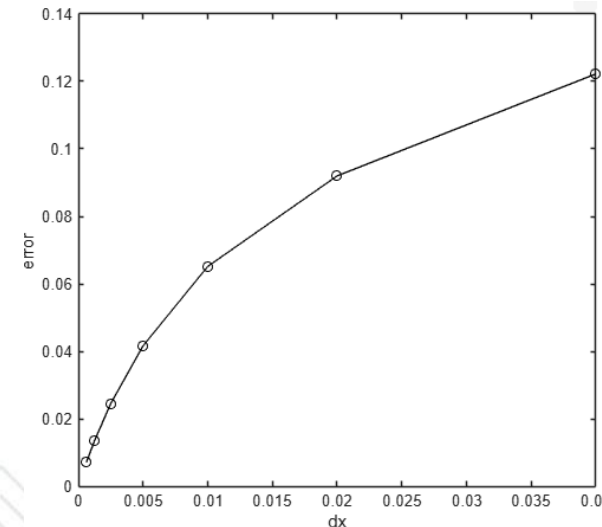
$$\epsilon = \frac{1}{N} \sum_i |u_i - u_{i,\text{exact}}|$$

Convergence



$$\epsilon = \frac{1}{N} \sum_i |u_i - u_{i,\text{exact}}|$$

- Calculate the solution, then calculate the error at some set time
- Repeat this for increasing values of N , and for each, recover an error ϵ_N that *should* decrease as N increases
 - Remember, the error better decrease, but we are specifically examining the rate of decrease (for EE FO BD, error is $O(\Delta x)$)
- Plot error vs. $dx = \frac{x_{\text{end}} - x_{\text{init}}}{N}$
- Is it obvious if this decreases at the correct rate?



Convergence

- Converting to log-log space works better

$$\epsilon \approx C \Delta x^p$$

$$\Delta x_1 \rightarrow \epsilon_1$$

$$\Delta x_2 \rightarrow \epsilon_2$$

$$\frac{\epsilon_1}{\epsilon_2} = \frac{C \Delta x_1^p}{C \Delta x_2^p}$$

$$\frac{\epsilon_1}{\epsilon_2} = \left(\frac{\Delta x_1}{\Delta x_2} \right)^p$$

$$\ln \left(\frac{\epsilon_1}{\epsilon_2} \right) = \ln \left[\left(\frac{\Delta x_1}{\Delta x_2} \right)^p \right]$$

$$\underbrace{\ln \left(\frac{\epsilon_1}{\epsilon_2} \right)}_y = p \underbrace{\ln \left(\frac{\Delta x_1}{\Delta x_2} \right)}_x + b$$

0

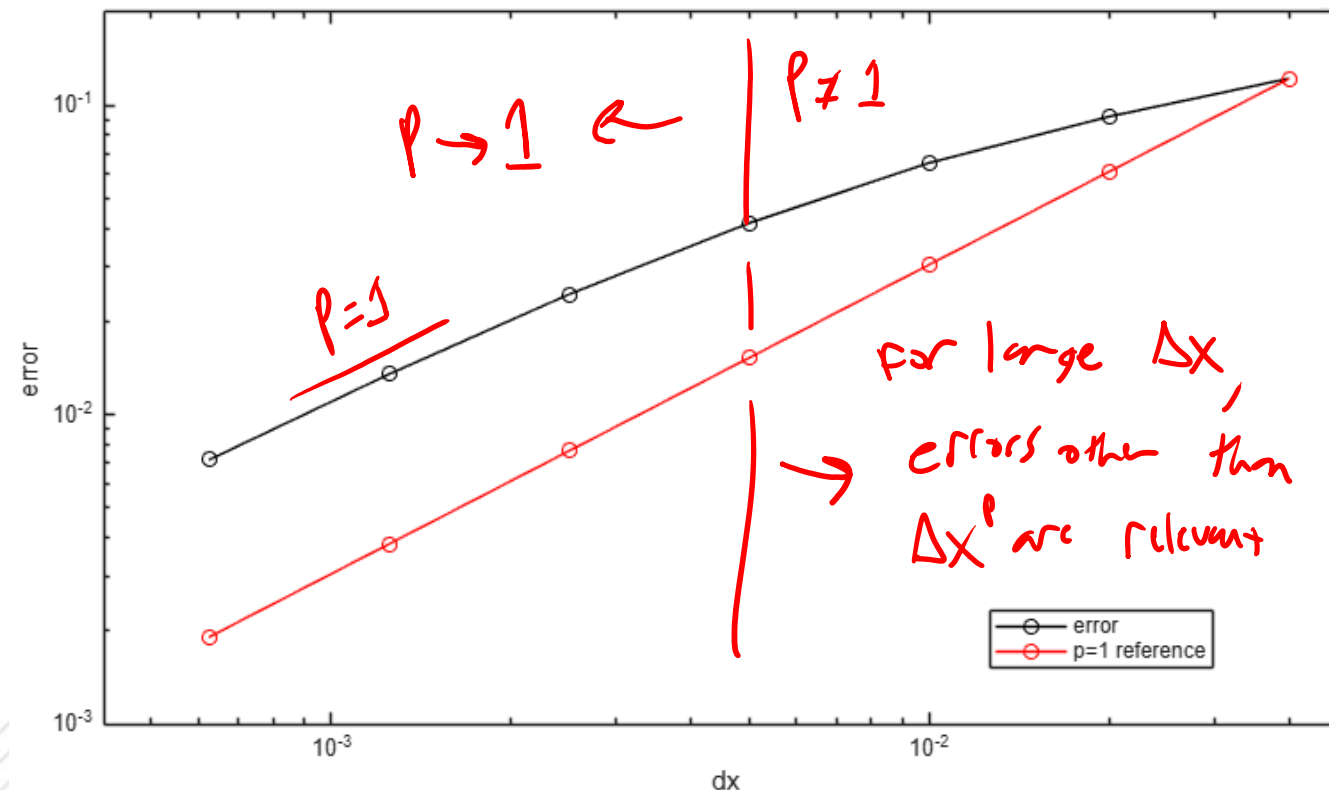
- The slope in log-log space should equal the power of Δx

Convergence

Note assignment #1 doesn't ask you to plot in log-log space or confirm convergence rate, but feel free to explore.

- We usually plot the data, and a curve (line with slope p in log-log space) next to our data to confirm convergence rate

- $\epsilon_{ref} = \epsilon_0 \left(\frac{\Delta x}{\Delta x_0} \right)^p$
- Generally, we consider visual confirmation sufficient to establish correct convergence



Convergence

- Showing that your code converges at the expected rate is an important part of verification because it supports the idea that you implemented everything correctly
- While we've explored convergence in space, you can do the same thing for $\Delta t \rightarrow 0$ and verify temporal convergence
 - Recall convergence for time and space can be different depending on the utilized schemes
- Note: formalizing convergence requires adequately smooth solutions. This would not work well for e.g., a square pulse.

Conservation

- We often verify our codes by exploring conservation, that is, the ability of our solution to conserve quantities that should physically be conserved (e.g., mass, momentum, etc.)

- Consider the linear advection equation $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$ with periodic boundary conditions. What must be true about the volume integral of u over the domain?

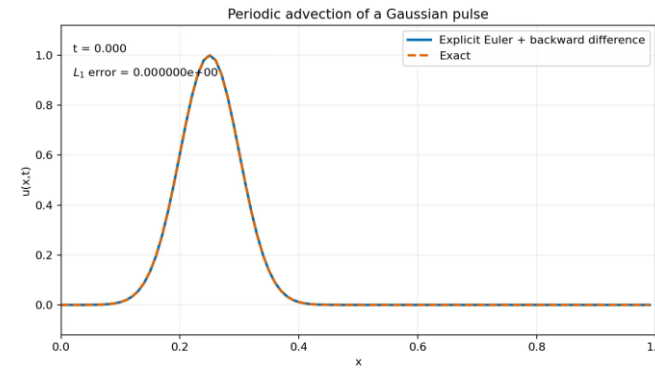
$$\int_V \frac{\partial u}{\partial t} dV + \int_V a \frac{\partial u}{\partial x} dV = 0 \quad \xrightarrow{\text{divergence theorem}} \quad \int_V \nabla \cdot \vec{F} dV = \int_A \vec{F} \cdot \hat{n} dA$$

$$\frac{\partial}{\partial t} \int_V u dV + a \int_V \frac{\partial u}{\partial x} dV = 0 \quad \xrightarrow{\text{per } \Delta x} \quad \text{flow out} = \text{flow in}$$

- Total amount of u is unchanged over time

$$\frac{\partial}{\partial t} \int_V u dV = 0$$

Conservation



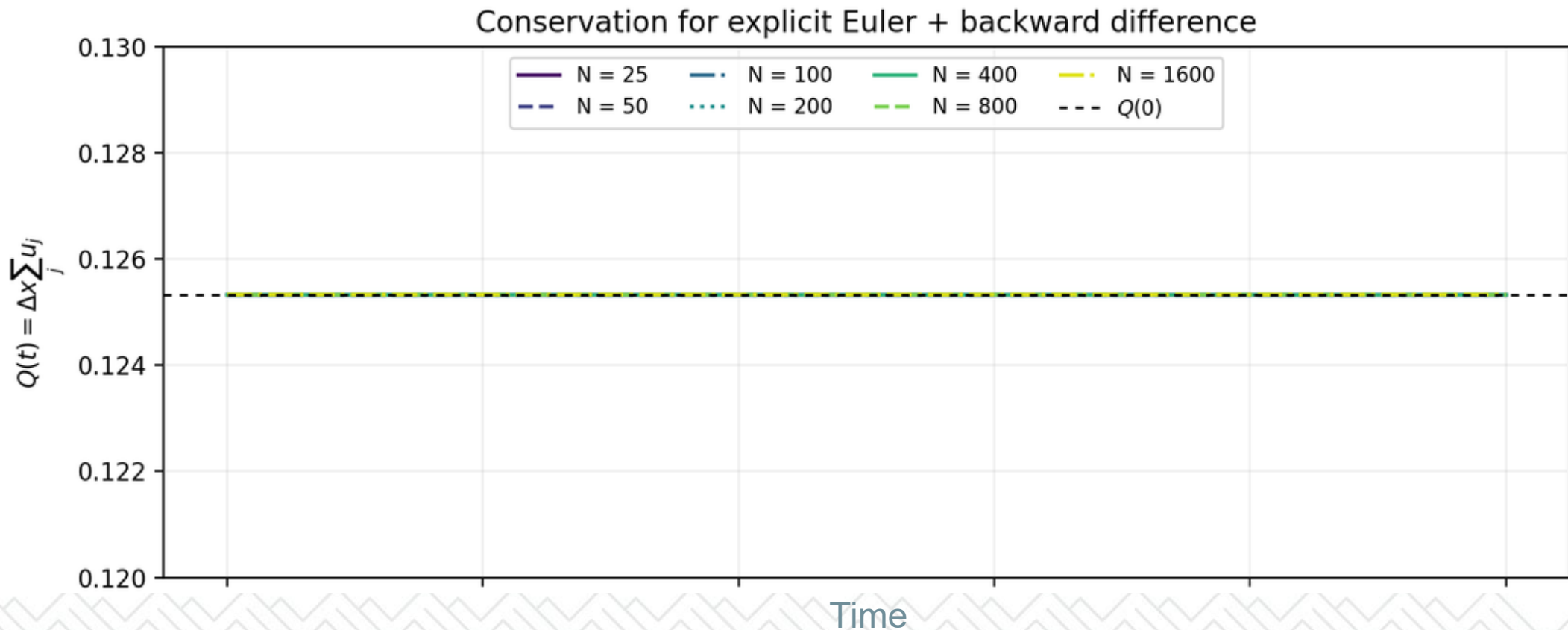
- Integrate over the total amount of “stuff” you want to conserve and evaluate how it changes over time
- Since u should be conserved for linear advection, let's check:

$$Q(t) = \int u dx \approx \sum_i u_i(t) \Delta x$$

- The answer is expected to vary with schemes and mathematical formulations

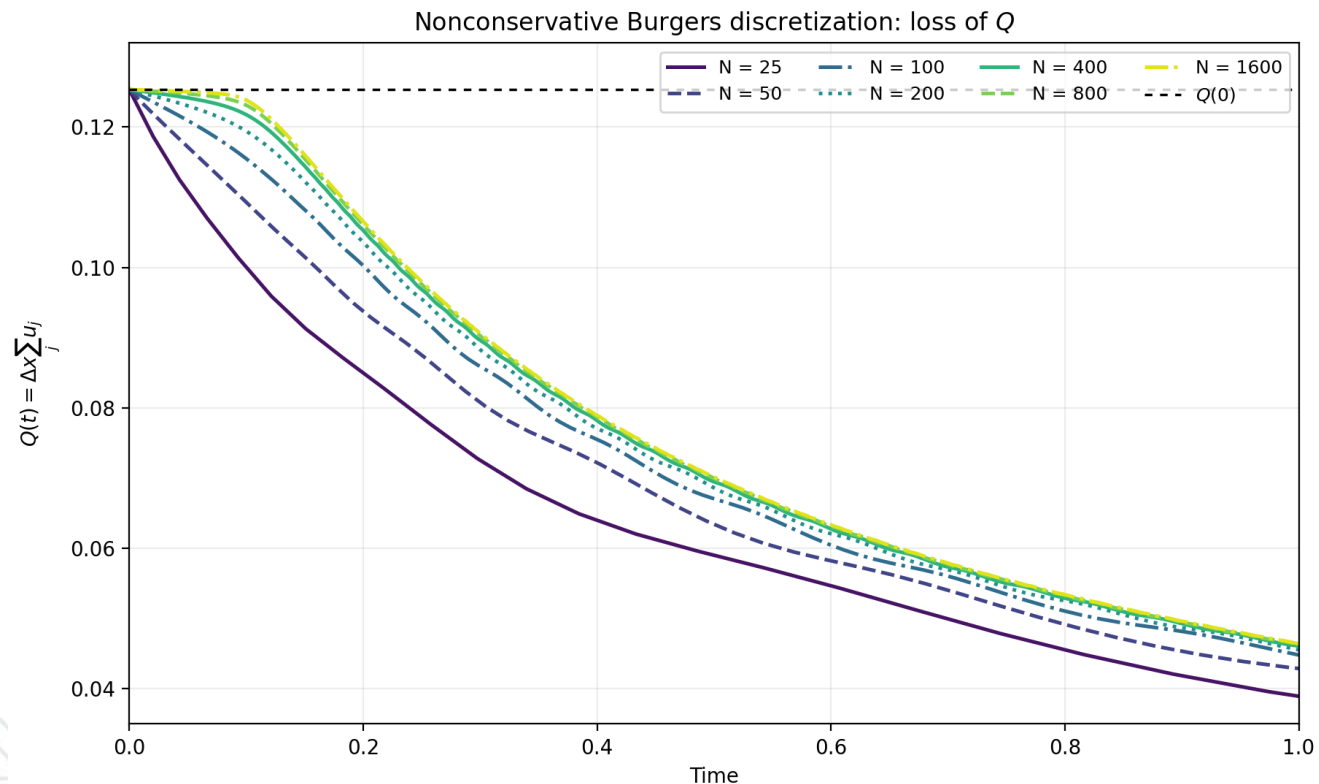
Conservation

- EE FO BD is naturally conserving for linear advection (Good!)



Conservation

- Let's explore EE FO BD for the non-linear Burgers equation
- Mathematically u should also be conserved (in any form)
- $$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$$
- This equation/scheme are not conservative, we constantly lose u
- Higher resolutions help, but don't fix the overall problem

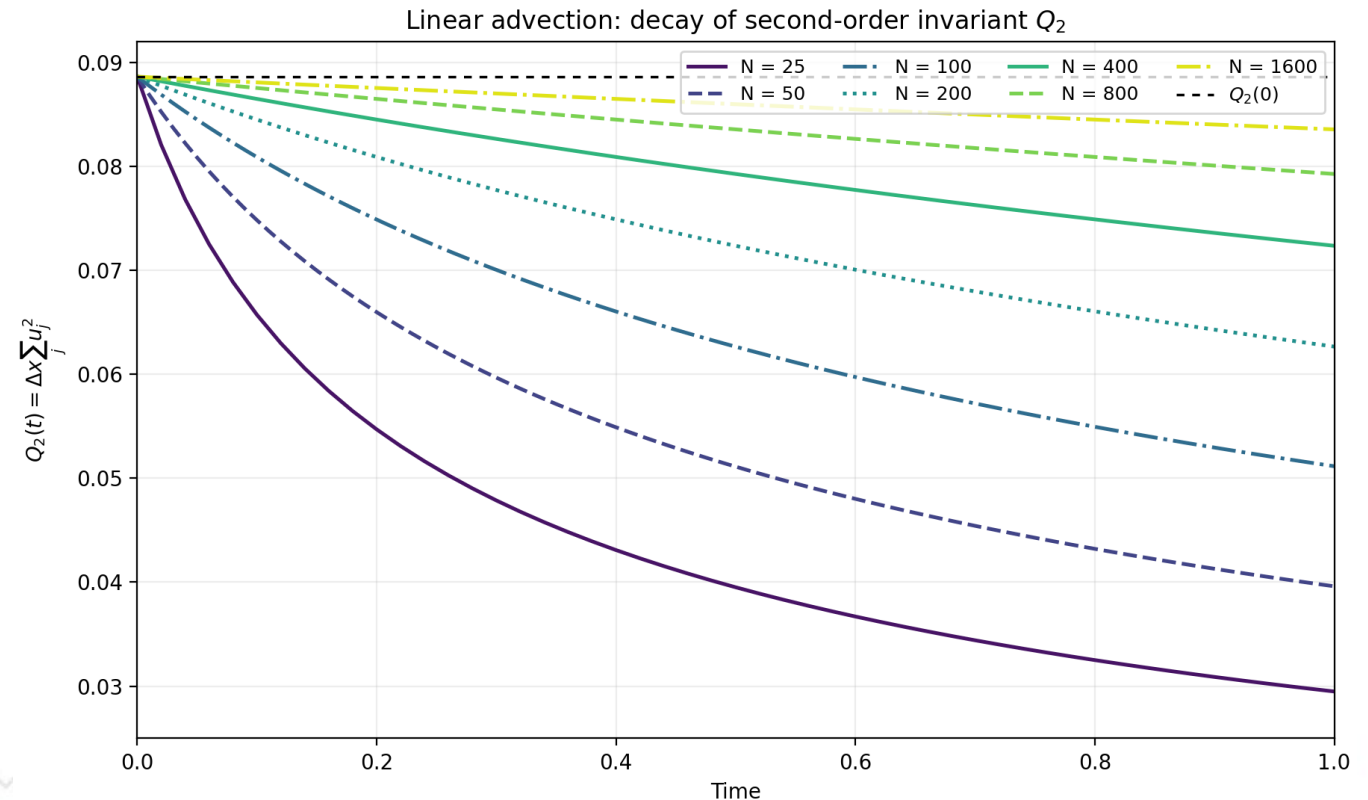


Conservation

- We just explored “primary” or “mass” conservation
- We can also explore “secondary” or “energy” conservation defined as $Q_2(t) = \int u^2 dx \approx \sum_i u_i^2(t) \Delta x$
 - We call it “energy” conservation because it resembles kinetic energy
- We can (but won’t) mathematically show this quantity should be conserved for linear advection with periodic boundaries

Conservation

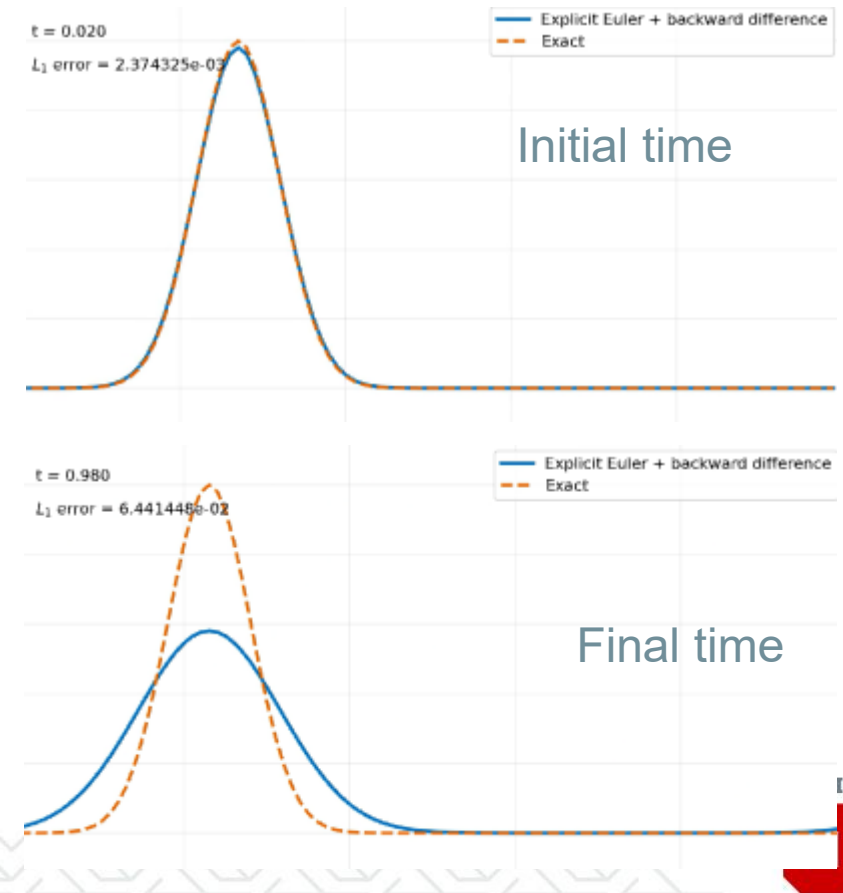
- EE FO BD of linear advection equation, secondary conservation
- Even though this equation and scheme passed primary conservation, they failed secondary conservation
- Grid resolution plays a much stronger role here



Conservation

- Why is this happening for secondary conservation?
- Note that this is a dissipative scheme, it makes the solution look like it diffuses
- Distribution of energy is very important

	Initial time			Final time		
	0	10	0	3	4	3
$\sum u$:	10			$3+4+3 = 10$		
$\sum u^2$:	100			$9+16+9 = 34$		



Conservation

- Many schemes may satisfy primary conservation, but not secondary conservation
- However, secondary conservation is important to kinetic energy in fluid mechanics
- Sometimes, we sacrifice secondary conservation for other numerically advantageous properties

Validation

- Once you are confident that you implemented your equations correctly and that they are doing what you expect, you can begin to focus on validation
- When validating, we care about:
 - Ensuring the correct governing equations are being solved
 - Identifying that important physics are being properly recovered
 - Grounding our answers in theory or measurements
 - Note that these can be experimental, or sometimes very high-fidelity simulations

Validation

- Consider the following: You implement the full three-dimensional Euler equations in a solver. You make sure your solution to a test problem converges properly, is stable, and you've shown great conservation properties. You have access to a supercomputer, so you refine to a very small grid and small timesteps that should be able to capture physics at all scales. You then use this code to calculate friction drag over an airplane wing. The results come back showing zero drag.
- What happened?

Validation

- When using CFD, your equations must include the required physics for the problem you wish to simulate
- For example:
 - If boundary layers are important, you must have viscous effects
 - If simulating high-speed flows, you must use compressible equations
 - If predicting mass transport, you must include species transport
 - If reactions are important, you must have chemical source terms
- A major part of validation is ensuring you do not miss physics

Validation

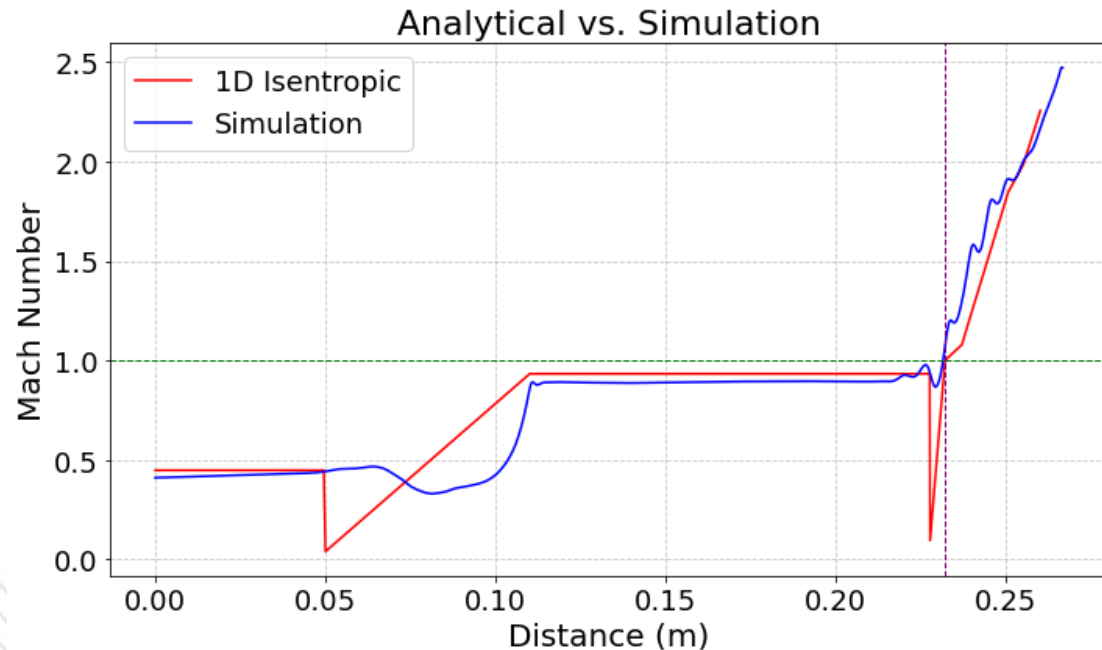
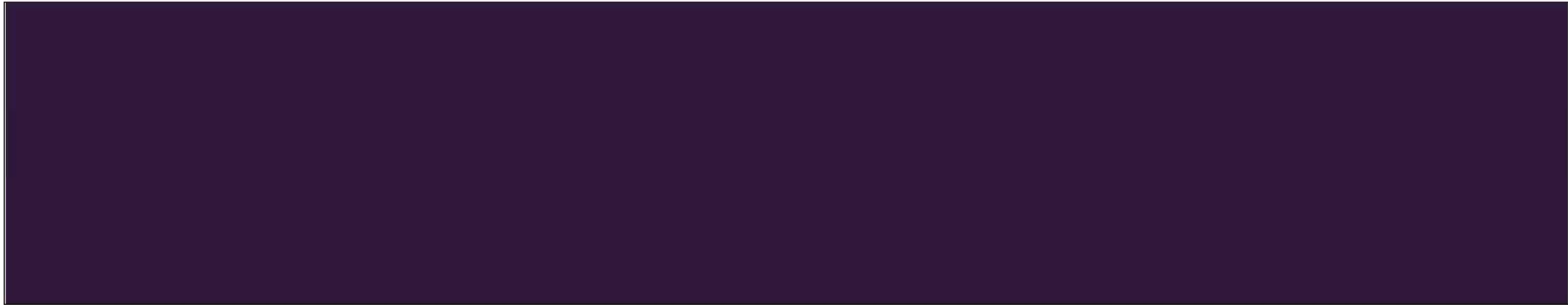
- In practice, we want test cases that we can validate on
 - Theoretical predictions
 - Experimental measurements
 - High-fidelity simulation results
- If you do not at least have an intuition on whether your results are reasonable, you must perform a validation comparison
 - Even if you're an expert, validation is still likely beneficial

Validation

- Do any of the equations you've learned in courses (or can find online/in a textbook) describe at least part of your problem? Are the answers you get from those equations close to your simulation results?
 - Even when data is available, back-of-the-napkin calculations are great
- If you search through research papers, NASA databases, etc., can you find results from a similar problem? Use your code to solve that problem and compare results directly.

Validation

Example: Cold flow through a hybrid rocket

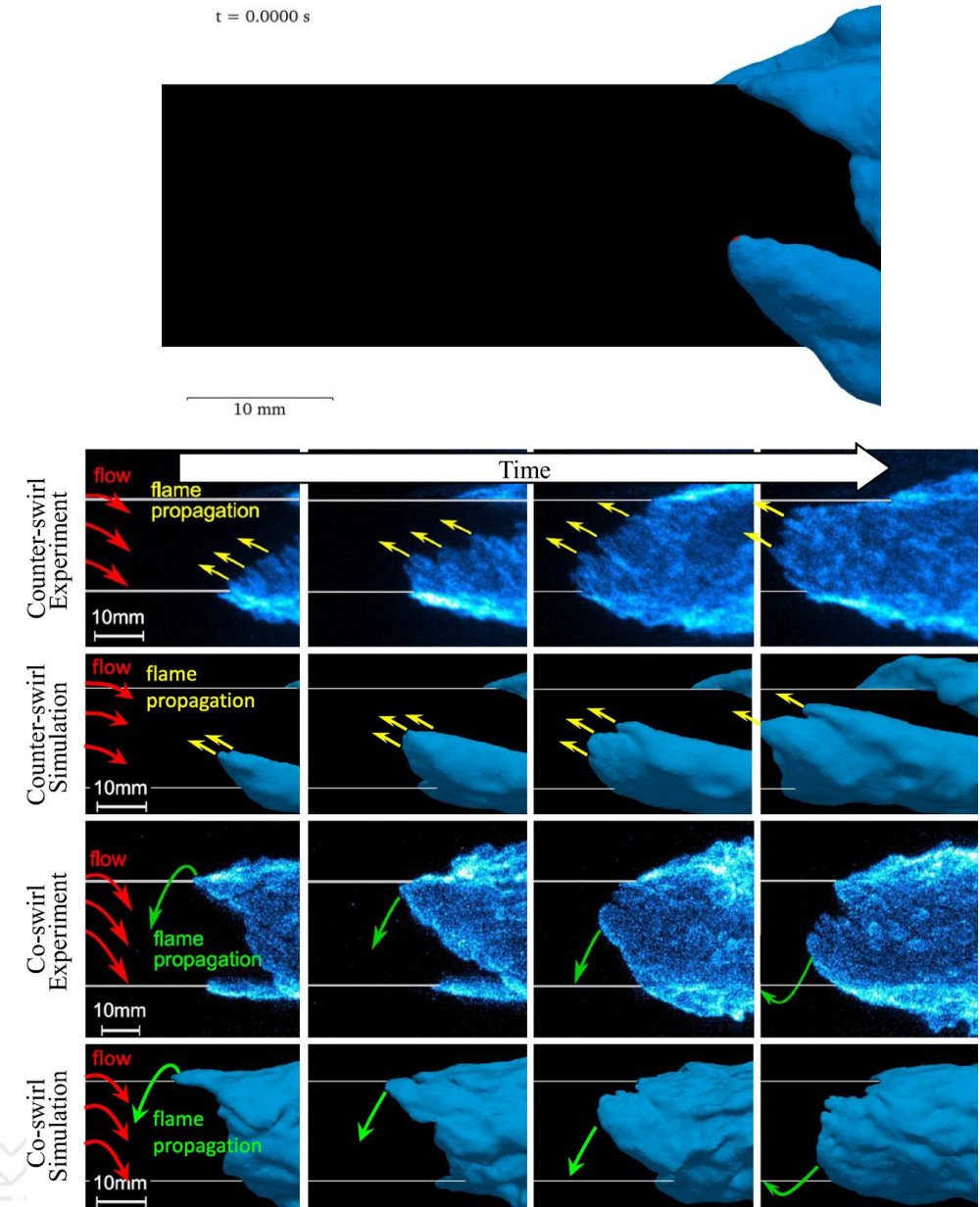
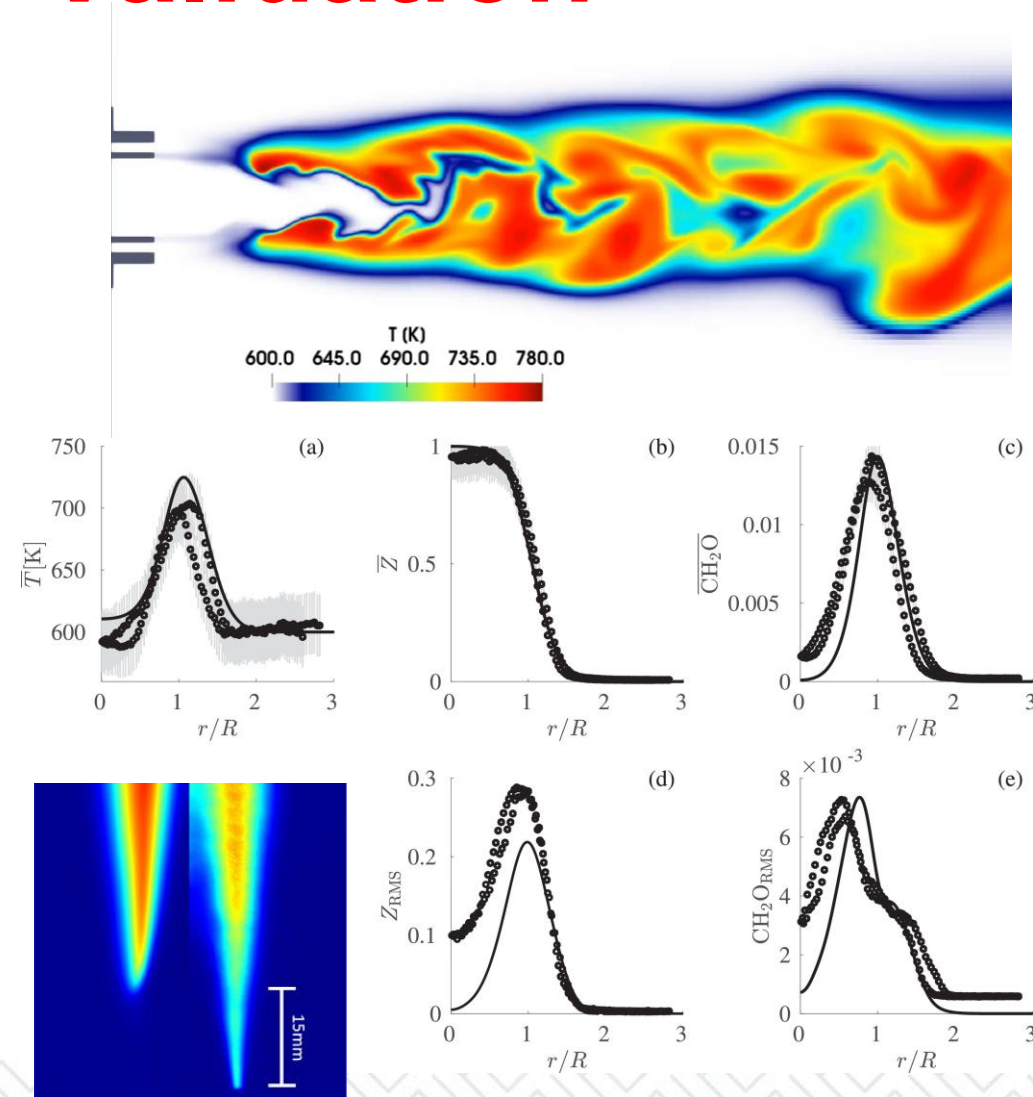


$$\frac{A_2}{A_1} = \frac{M_1}{M_2} \left[\frac{1 + \frac{\gamma-1}{2} M_1^2}{1 + \frac{\gamma-1}{2} M_2^2} \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

- This was based on an experiment, so we were also able to compare to obtained thrust measurements

Example: Reacting flows vs. experiments

Validation



Looking Forward

- To validate compressible flows, we should understand the physics behind compressible flows
- The next few lectures will be introductions to compressible flows, including isentropic flows and shockwaves
- Note: This will be very similar to what I teach about compressible flows in Aerospace Propulsion